PROJECT REPORT

FOR

Foundations of Artificial Intelligence

AI-Powered Team Formation Tool

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report " **AI-Powered Team Formation Tool** "
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ABSTRACT

Team formation is an important task in academic institutions and project-based learning environments, where individuals with different skills and abilities must work together to achieve a common goal. Traditional team formation is usually performed manually, which can be time-consuming and may result in an unequal distribution of skills, responsibilities, and workload among team members. To overcome these challenges, the **AI-Powered Team Formation Tool** is proposed to automate the process of creating balanced, effective, and suitable teams based on predefined criteria.

The proposed system considers various participant attributes, including technical skills, attendance, portfolio submission, and leadership experience, to identify suitable members for team formation. It uses a **Forward-Chaining Rule Engine** to evaluate predefined rules and determine participant eligibility and role suitability. Furthermore, the system applies the **Constraint Satisfaction Problem (CSP)** approach along with a backtracking algorithm to generate teams that satisfy requirements such as team size, skill diversity, and role distribution. These techniques help the system identify valid team combinations while considering the specified constraints.

The system also incorporates a fairness evaluation module to assess the balance of skills and responsibilities among the generated teams. It identifies constraints that cannot be satisfied and provides information about the limitations encountered during team formation.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO.
	ABSTRACT	3
1	INTRODUCTION	5
2	LITERATURE REVIEW	6
3	PROPOSED SYSTEM	10
	3.1ARCHITECTURE DIAGRAM	11
4	MODULES DESCRIPTION	12
	4.1 MODULE 1	12
	4.2 MODULE 2	12
	4.3 MODULE 3	12
	4.4 MODULE 4	13
	4.5 MODULE 5	13
	4.6 MODULE 6	13
	4.7 MODULE 7	13
5	IMPLEMENTATION AND RESULTS	14
	5.1EXPERIMENTAL SETUP	14
	5.2RESULTS	15
6	CONCLUSION AND FUTURE WORK	17
7	REFERENCES	19

CHAPTER 1

INTRODUCTION

Team formation is an important process in academic institutions and project-based environments, where individuals with different skills and abilities work together to achieve a common goal. Traditional team formation is usually performed manually, which can be time-consuming and may result in an unequal distribution of skills and responsibilities among team members. Therefore, an automated approach is needed to create balanced and effective teams.

The **AI-Powered Team Formation Tool** aims to automate team formation based on participants' technical skills, attendance, portfolio submission, and leadership experience. The system uses a Forward-Chaining Rule Engine to identify eligible participants and a Constraint Satisfaction Problem (CSP) approach with backtracking to generate teams that satisfy predefined requirements, including team size, skill diversity, and role distribution.

The system also evaluates team fairness, identifies unmet constraints, and recommends suitable campus meeting points. Developed using Python and Streamlit, the proposed tool provides a user-friendly interface for generating and viewing teams. It aims to reduce manual effort, improve fairness, and support effective collaboration among team members in educational institutions and project-based environments.

In addition, the proposed system helps improve transparency and consistency in the team formation process by applying the same predefined rules and constraints to all participants. It enables users to understand how teams are formed and how roles are assigned based on individual capabilities. The system can be further enhanced by incorporating machine learning techniques to predict team performance and recommend better team combinations. Thus, the AI-Powered Team Formation Tool provides a flexible and efficient approach to managing team formation and promoting balanced participation in collaborative projects.

CHAPTER 2
LITERATURE REVIEW

Paper 1: USTF: A Unified System of Team Formation

Wang, J. et al. (2016).

"USTF: A Unified System of Team Formation."

This paper focuses on automated team formation by considering individual skills and team requirements. It provides a foundation for selecting suitable members to perform tasks efficiently. This research is relevant to our project in developing an automated system that forms teams based on participants' skills and predefined requirements.

Paper 2: Parameterized Complexity of Team Formation in Social Networks

Bredereck, R. et al. (2018).

"Parameterized Complexity of Team Formation in Social Networks."

This paper investigates the computational challenges involved in forming teams within social networks. It explores algorithmic approaches to identify suitable team members while considering task requirements and relationships between individuals. This research is relevant to our project in understanding the complexity of team formation and developing efficient team allocation algorithms.

Paper 3: A Unified Framework for Effective Team Formation in Social Networks

Selvarajah, K., Moradian Zadeh, P., Kobti, Z., Palanichamy, Y., & Kargar, M. (2021).

"A Unified Framework for Effective Team Formation in Social Networks."

This paper proposes a framework for forming effective teams by considering expertise, communication costs, trust, and collaboration requirements. It uses a multi-objective optimization approach to identify suitable team combinations. This research is relevant to our project in evaluating multiple factors during team formation.

Paper 4: In Search of Diverse and Connected Teams

Gómez-Zará, D., Das, A., Pawlow, B., & Contractor, N. (2022).

"In Search of Diverse and Connected Teams: A Computational Approach to Assemble Diverse Teams Based on Members' Social Networks."

This paper presents a computational approach for forming diverse teams while considering the relationships among members. It uses a multi-objective optimization algorithm to balance team diversity and familiarity. This research supports our project's objective of distributing different skills among team members to create balanced teams.

Paper 5: Social Structure Optimization in Team Formation

(2016).

"Social Structure Optimization in Team Formation."

This paper introduces an optimization framework that considers social relationships when forming teams. It explores how team structure and relationships between members can influence team outcomes. This research is relevant to our project in improving team composition and considering factors beyond individual skills.

Paper 6: A Branch-and-Bound Algorithm for Team Formation on Social Networks

Berktaş, N., & Yaman, H. (2020).

"A Branch-and-Bound Algorithm for Team Formation on Social Networks."

This paper presents an algorithm for identifying suitable teams while minimizing communication costs. It uses branch-and-bound techniques to solve team formation problems efficiently. This research is relevant to our project because it demonstrates how algorithmic optimization can be used to find valid team combinations while satisfying predefined constraints.

Paper 7: SoC-Constrained Team Formation with Self-Organizing Mechanism in Social Networks

Shi, Y., Peng, Z., Hong, L., & Yu, Q. (2017).

"SoC-Constrained Team Formation with Self-Organizing Mechanism in Social Networks."

This paper focuses on forming teams while considering required skills, communication costs, team size, and leadership responsibilities. It introduces algorithms for selecting members and assigning leadership roles. This research is relevant to our project in assigning suitable roles and maintaining appropriate team structures.

Paper 8: A Novel State Space Reduction Algorithm for Team Formation in Social Networks

(2021).

"A Novel State Space Reduction Algorithm for Team Formation in Social Networks."

This paper proposes an algorithm that reduces the search space involved in identifying suitable team members. It considers skills and communication costs to improve the efficiency of team formation. This research is relevant to our project because it highlights the importance of efficient search techniques when generating teams that satisfy multiple constraints.

Paper 9: Efficient Algorithms for Team Formation with a Leader in Social Networks

Juang, M. C., Huang, C. C., & Huang, J. L.

"Efficient Algorithms for Team Formation with a Leader in Social Networks."

This paper investigates methods for selecting team members and identifying an appropriate leader based on project skill requirements and communication costs. It proposes algorithms to reduce the computational effort required to find suitable

teams. This research is relevant to our project in developing role assignment mechanisms and selecting team members based on their capabilities.

Paper 10: Assigning or Recommending You a Team? The Algorithmic Effects on Team Formation and Performance

Gómez-Zará, D. (2024).

"Assigning or Recommending You a Team? The Algorithmic Effects on Team Formation and Performance."

This study examines how team formation algorithms influence the selection of team members, collaboration patterns, and team composition. It explores how different algorithmic approaches can affect team formation decisions. This research is relevant to our project in understanding the role of automated recommendations in improving team selection and supporting effective collaboration

CHAPTER 3

PROPOSED SYSTEM

The proposed **AI-Powered Team Formation Tool** is designed to automate the process of forming balanced and effective teams based on participants' skills, attendance, portfolio submission, and leadership experience. The system aims to overcome the limitations of manual team formation, which can be time-consuming and may result in an unequal distribution of skills and responsibilities. By applying rule-based reasoning and constraint-based techniques, the proposed system generates suitable teams according to predefined requirements.

Initially, the system collects participant information, including technical skills, attendance details, portfolio submission status, and leadership experience. The collected data is validated and processed to ensure that it is suitable for team formation. A **Forward-Chaining Rule Engine** applies predefined rules to identify eligible participants and determine their suitability for different roles. This helps ensure that team formation follows consistent criteria.

The system then uses the **Constraint Satisfaction Problem (CSP)** approach to generate teams that satisfy requirements such as team size, skill diversity, and role distribution. A backtracking algorithm searches for valid team combinations and revisits earlier selections when the current combination does not satisfy the specified constraints. This approach helps identify suitable team arrangements while considering the requirements defined by the user.

A fairness evaluation module examines the generated teams by considering skill distribution, role allocation, and team balance. It also identifies constraints that cannot be satisfied and displays the relevant information to the user. Additionally, the system can recommend suitable campus meeting points to support collaboration among team members. These features help users understand the generated results and make informed decisions about team composition.

3.1 ARCHITECTURAL DIAGRAM

The architecture of the proposed system can be represented as follows:

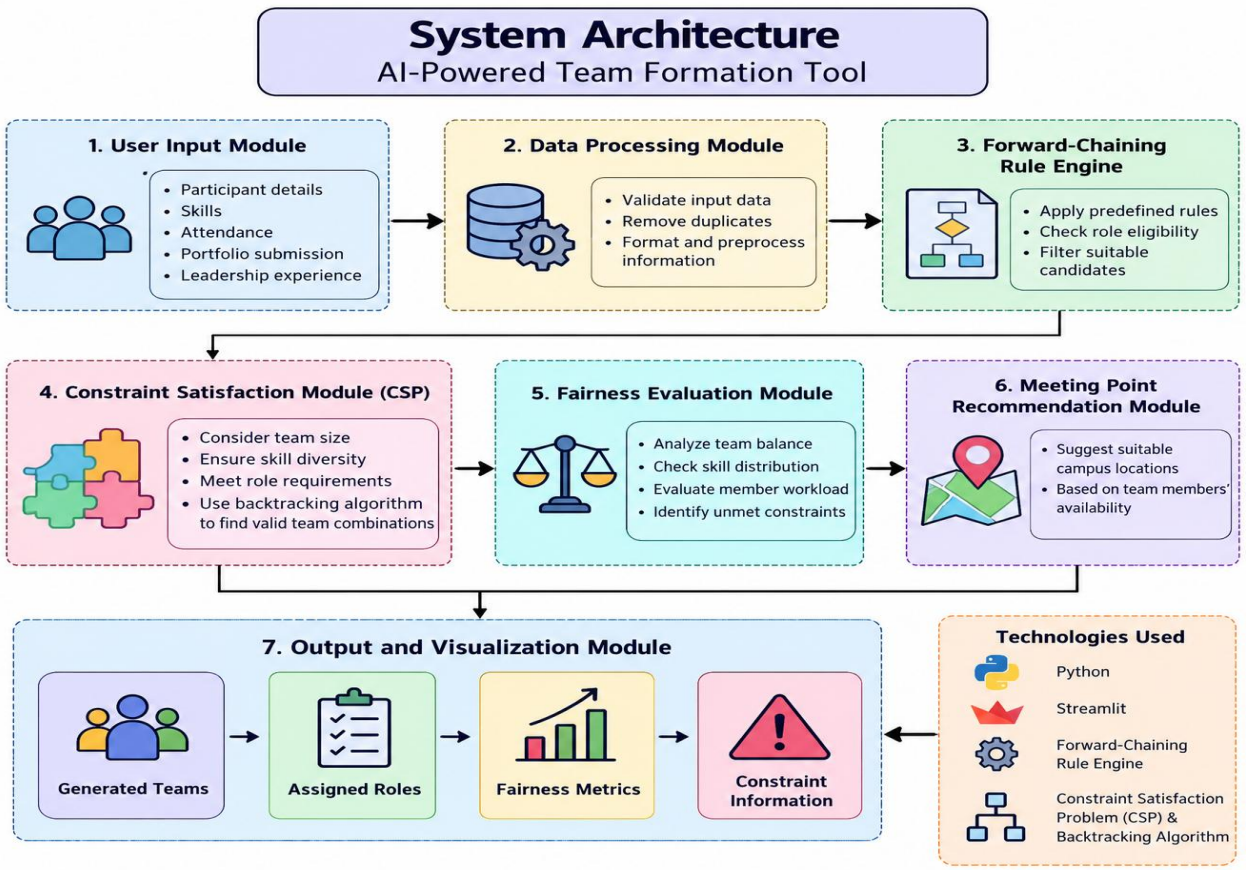


Figure 3.1: System Architecture of the AI-Powered Team Formation Tool

Figure 3.1 illustrates the system architecture of the AI-Powered Team Formation Tool. It shows the flow of information from user input and data preprocessing to rule-based processing and team formation using CSP and backtracking algorithms. The generated teams are evaluated for fairness, skill distribution, and role balance. Finally, the system displays the team formation results and stores relevant information in the database.

CHAPTER 4

MODULES DESCRIPTION

The AI-Powered Team Formation Tool consists of several modules that work together to generate balanced and effective teams based on predefined requirements. Each module performs a specific function, from collecting participant information to generating and evaluating the final teams.

4.1 User Input Module

The User Input Module collects the necessary information about participants, including their names, technical skills, attendance, portfolio submission, and leadership experience. It also allows users to specify team formation requirements, such as team size and required roles. These details serve as the input for the team formation process.

4.2 Data Preprocessing Module

The Data Preprocessing Module validates and organizes the collected participant information. It checks for missing values, duplicate records, and inconsistent data to improve the accuracy of subsequent processing. The validated information is then passed to the rule engine for further analysis.

4.3 Rule Engine Module

The Rule Engine Module uses a forward-chaining approach to apply predefined rules to participant data. It checks participant eligibility based on attendance, portfolio submission, skills, and leadership experience. The module also identifies suitable participants for different team roles according to the specified requirements.

4.4 Team Formation Module

The Team Formation Module generates teams using the Constraint Satisfaction Problem (CSP) approach and a backtracking algorithm. It considers constraints such as team size, skill diversity, and role distribution to identify suitable team combinations. The backtracking algorithm explores possible combinations and revises selections when the constraints are not satisfied.

4.5 Fairness Evaluation Module

The Fairness Evaluation Module assesses the quality of the generated teams by examining skill distribution, role allocation, and team balance. It identifies differences between teams and highlights constraints that cannot be satisfied. This module helps users understand whether the generated teams meet the specified requirements.

4.6 Meeting Point Recommendation Module

The Meeting Point Recommendation Module suggests suitable campus locations where team members can meet and collaborate. Recommendations can be based on available campus locations and team requirements. This feature aims to support convenient communication and coordination among team members.

4.7 Output and Visualization Module

The Output and Visualization Module presents the final results through a user-friendly interface. It displays the generated teams, assigned roles, fairness evaluation, and unmet constraints. It can also display meeting-point recommendations. The module helps users understand the results and review the generated team arrangements easily.

CHAPTER 5

IMPLEMENTATION AND RESULTS

The AI-Powered Team Formation Tool collects participant information, processes the data, and generates balanced teams based on predefined requirements. The system considers technical skills, attendance, portfolio submission, and leadership experience to identify suitable team members. It applies rule-based reasoning and constraint satisfaction techniques to generate teams, assign roles, evaluate fairness, and identify unmet constraints. The results are presented through a user-friendly interface, helping users understand the team formation process.

5.1 EXPERIMENTAL SETUP

The system is designed using Python as the backend programming language and Streamlit for the frontend interface. A database such as SQLite can be used to store participant details and generated team information. The Forward-Chaining Rule Engine applies predefined rules to determine participant eligibility and role suitability. The Constraint Satisfaction Problem (CSP) approach and backtracking algorithm are used to generate teams that satisfy requirements such as team size, skill diversity, and role distribution. The fairness evaluation module examines team balance and identifies constraints that cannot be satisfied. The system can also incorporate a meeting-point recommendation module to suggest suitable campus locations for team collaboration. The implementation combines rule-based decision-making and classical search algorithms without requiring machine-learning or deep-learning models.

5.2 RESULTS

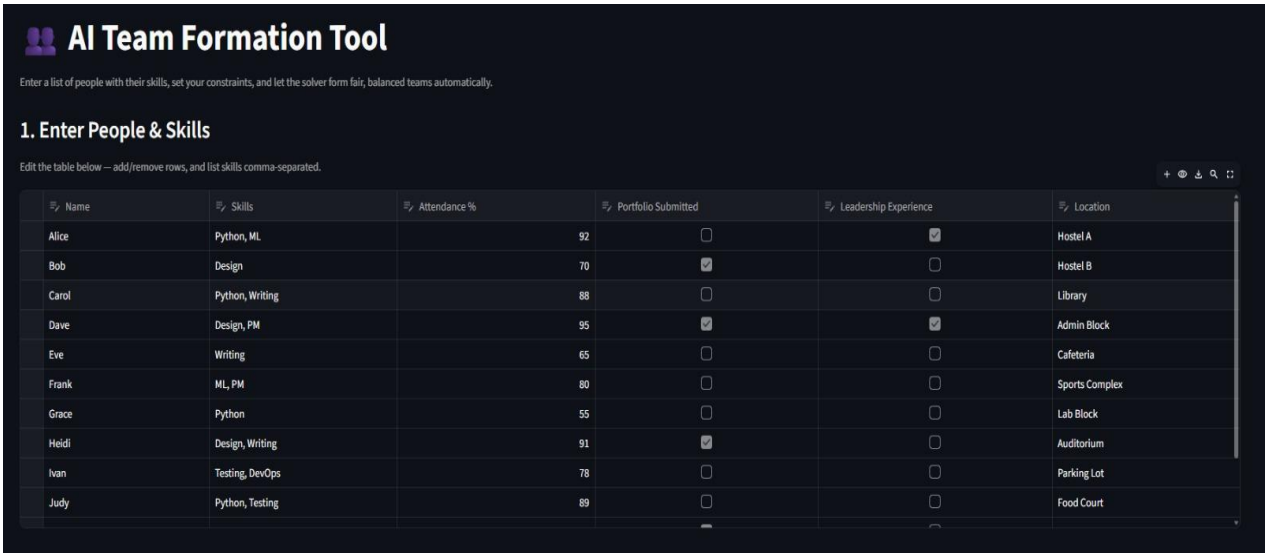


Figure 5.1: User Interface of the AI-Powered Team Formation Tool

Figure 5.1 shows the user interface of the AI-Powered Team Formation Tool, where participant details such as name, skills, attendance percentage, portfolio submission, leadership experience, and location are entered. This information is used to evaluate participants and support the automatic formation of balanced teams.



Figure 5.2: Participant Eligibility Verification Interface

Figure 5.2 illustrates the participant eligibility verification interface, which uses backward-chaining reasoning to determine whether a selected participant is eligible for a specific role. The user selects a participant and enters the target role, and the system evaluates the relevant predefined rules to generate a proof of eligibility. This helps explain the reasoning behind role assignments and supports transparent, rule-based team formation.

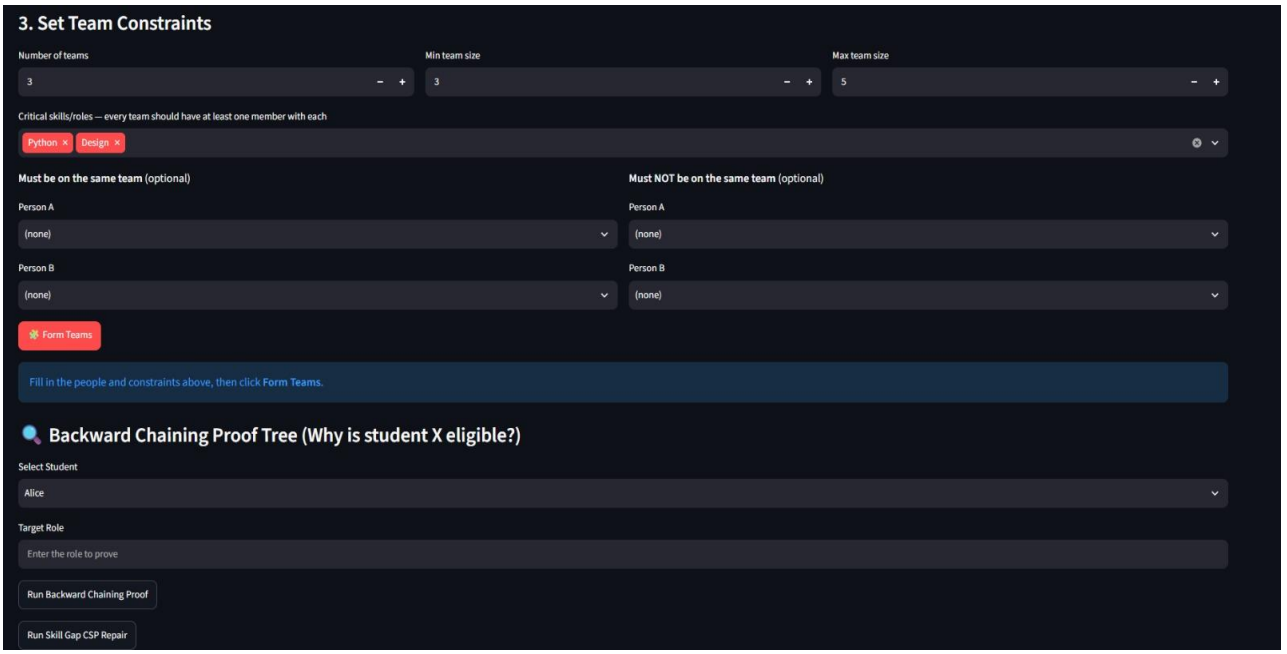


Figure 5.3: Team Constraint Configuration and Rule-Based Processing

Figure 5.3 illustrates the team constraint configuration interface, where users specify the number of teams, minimum and maximum team sizes, critical skills, and conditions for grouping participants. It also includes backward-chaining proof and CSP-based skill gap repair options to verify role eligibility and identify suitable team combinations based on predefined constraints.

CHAPTER 6

FUTURE WORK AND CONCLUSION

6.1 FUTURE WORK

The AI-Powered Team Formation Tool can be enhanced in several ways to improve its efficiency, accuracy, and usability. In the future, machine learning techniques can be integrated to predict team performance based on participants' skills, experience, and previous project outcomes. Advanced optimization algorithms can be introduced to improve skill distribution and generate more balanced teams. The system can also incorporate real-time availability tracking to simplify team scheduling and coordination. Integration with college management systems can automate participant data collection and reduce manual effort. Furthermore, a mobile application and cloud-based deployment can improve accessibility. The meeting-point recommendation feature can be enhanced by considering team members' locations and preferences. These improvements can make the system more flexible, scalable, and useful for academic and professional environments.

6.1.1 Scope for Improvement

- Improve fairness in skill and workload distribution.
- Optimize algorithms for faster team formation.
- Enhance the user interface and result visualization.

6.1.2 Future Enhancements

- Integrate machine learning to predict team performance.
- Develop a mobile application and cloud-based platform.
- Add smart meeting schedule and location recommendations.

6.2 CONCLUSION

The AI-Powered Team Formation Tool provides a systematic approach to automating team formation based on participants' skills, attendance, portfolio submission, leadership experience, and predefined constraints. The system combines a Forward-Chaining Rule Engine with Constraint Satisfaction Problems (CSP) and backtracking techniques to identify suitable team combinations and assign appropriate roles. The fairness evaluation module helps assess skill distribution, team balance, and unmet constraints. The user-friendly interface developed using Python and Streamlit simplifies data entry and result visualization. Overall, the proposed system aims to reduce the effort involved in manual team formation, promote fairer participation, and support effective collaboration. With further development and testing, the tool can be extended to meet the team formation needs of educational institutions and project-based organizations.

CHAPTER 7

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